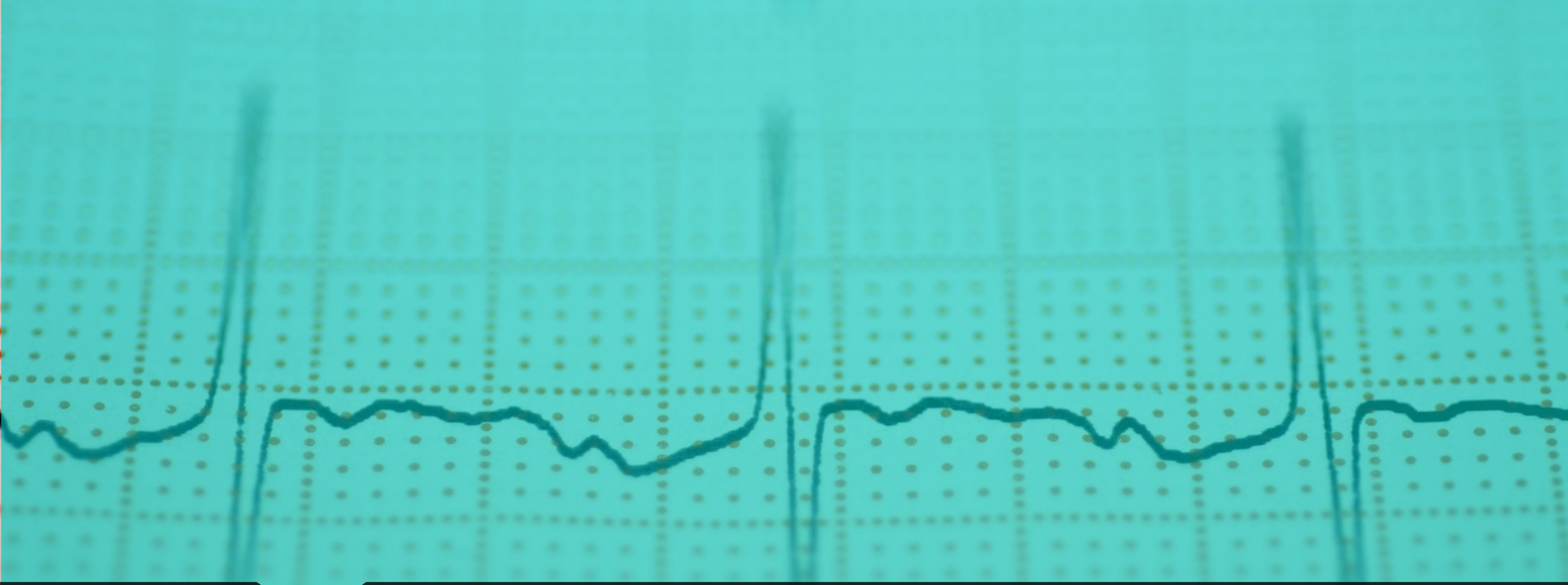
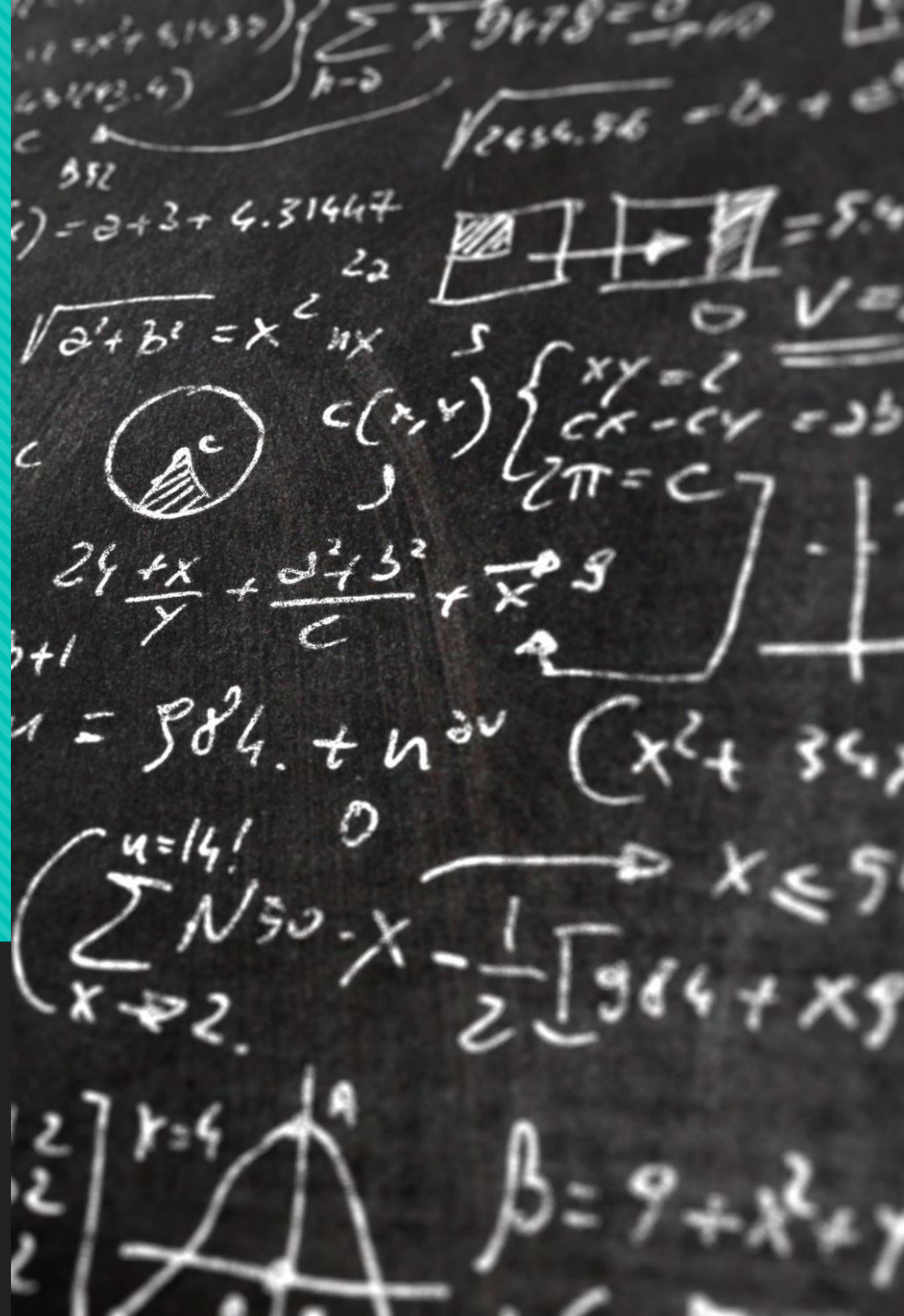


Project Proposal

Stress Detection and Analysis Using EEG and ECG Data on Azure

Project Type:

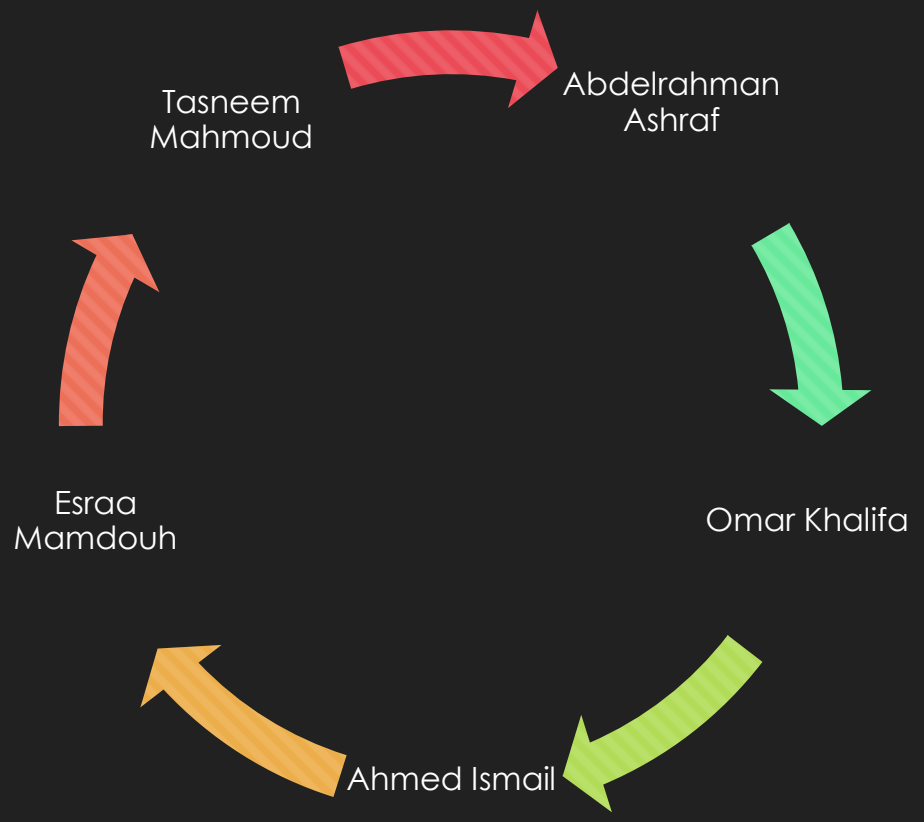
Data Engineering / Machine Learning Project



Project Purpose and Need

- Stress detection and analysis using biometric data is crucial for modern healthcare and wellness solutions. This project aims to classify stress levels based on EEG and ECG data, leveraging cloud-based infrastructure on Azure and deploying the final solution on Streamlit. The system will provide a scalable, accurate solution to monitor stress levels in real-time, potentially benefiting healthcare professionals and wellness applications.

Team Members



Data Sources:

- Kaggle Dataset: ECG and EEG Stress Features
- NCBI Article: ECG and EEG based detection and multilevel classification of stress using machine learning for specified genders: A preliminary study

Tools and Technologies:

Cloud: Azure (Blob Storage, Synapse Analytics, Data Factory, Machine Learning)

Programming & Libraries: Python (Pandas, NumPy, Scikit-learn)
Deployment: Streamlit

Visualization: Power BI
Machine Learning Integration: Azure Machine Learning, Jupyter Notebooks

Project Plan and Timeline

Week 1: Data Managemen t and Azure Storage Setup

Data Acquisition and Organization:
- Gather datasets from Kaggle and anonymize any sensitive data.

Azure Storage Setup:
- Set up Azure Blob Storage and organize data into /raw-data and /processed-data folders.

Data Exploration and Preprocessing:
- Perform exploratory data analysis (EDA), handle missing values, and clean data.
- Normalize or standardize features as necessary.

Deliverables:
- A well-organized dataset in Azure Blob Storage.
- EDA report and preprocessed data ready for modeling.

Week 2: Data Warehousing and SQL Setup

Azure Data Warehouse Implementation:
- Set up Azure Synapse Analytics or Azure SQL Database for efficient storage.

Data Integration:
- Use Azure Data Factory for data pipelines to move data to the data warehouse.

Deliverables:
- Functioning data warehouse with integrated data.
- SQL scripts for data verification.

Week 3: Machine Learning Model Development and Azure Integration

- Data Science and Model Development:
- Perform advanced data analysis and correlations related to stress levels.
- Develop machine learning models (SVM, kNN, etc.) to classify stress levels using the preprocessed data.

Azure Machine Learning Integration:
- Train, tune, and manage models in Azure ML Studio.

Deliverables:
- Trained ML models and documented performance metrics.

Week 4: Deployment, Real-Time Visualization, and Final Presentation

Model Deployment (Streamlit):
- Deploy the best-performing machine learning model on Streamlit for user interaction and inference.

Real-Time Data Visualization:
- Create real-time visualizations using Power BI and integrate it with the Streamlit app.

Final Report and Presentation:
- Prepare a comprehensive report covering the project lifecycle and present the final solution.

Deliverables:
- Deployed model on Streamlit, real-time dashboard, final report, and presentation.

TEAM MEMBER RESPONSIBILITIES

Team Member Responsibilities

- **Abdelrahman Ashraf:** Machine learning model development and Azure ML integration.
- **Omar Khalifa:** Data warehousing, Azure Synapse setup, and SQL queries.
- **Ahmed Ismail:** Data preprocessing, EDA, and Azure Data Factory pipelines.
- **Esraa Mamdouh:** Data acquisition, storage organization, and Power BI visualization.
- **Tasneem Mahmoud:** Cloud infrastructure setup on Azure and Streamlit deployment.

Instructor Reporting:

Every 4 days, the team leader (Abdelrahman Ashraf) will provide progress updates to the instructor, ensuring alignment with the project timeline and goals.